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AR Try-On Technologies Research Report

Objective

To research Augmented Reality (AR) technologies used in virtual try-on systems for clothes, shoes, jewellery, and accessories. The study focuses on tracking methods, implementation technologies, performance, and user experience.

1. Clothes Try-On

Technology Used

- * Body Tracking
- * Pose Estimation
- * Body Segmentation
- * Garment Overlay

Common Technologies

- * MediaPipe Pose
- * BlazePose
- * OpenPose

Working

The system detects body landmarks such as shoulders, elbows, hips, and knees. The clothing image or 3D garment is then aligned with these landmarks and displayed over the user's body in real time.

Advantages

- * Real-time visualization
- * Improved online shopping experience
- * Easy integration with mobile devices

Limitations

- * Inaccurate fitting for complex poses
- * Clothing deformation may appear unrealistic
- * Sensitive to lighting conditions



2. Shoe Try-On

Technology Used

- * Foot Tracking
- * Plane Detection
- * 3D Object Placement

Common Technologies

- * ARCore
- * ARKit
- * Snap AR

Working

The system identifies the user's feet and overlays a 3D shoe model. The model follows foot movement and orientation in real time.

Advantages

- * Highly interactive experience
- * Realistic product visualization
- * Useful for e-commerce applications

Limitations

- * Requires clear visibility of feet
- * Tracking accuracy decreases under poor lighting



3D



AR



Converse Chuck Taylor All Star -
Darkly Jaded



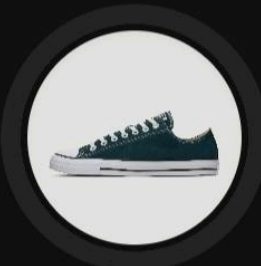
3D



AR



Converse Chuck Taylor All Star - Darkly Jaded



3. Jewellery Try-On

Technology Used

- * Face Tracking
- * Neck Tracking
- * Wrist Tracking

Applications

- * Necklaces
- * Earrings
- * Rings
- * Bracelets

Working

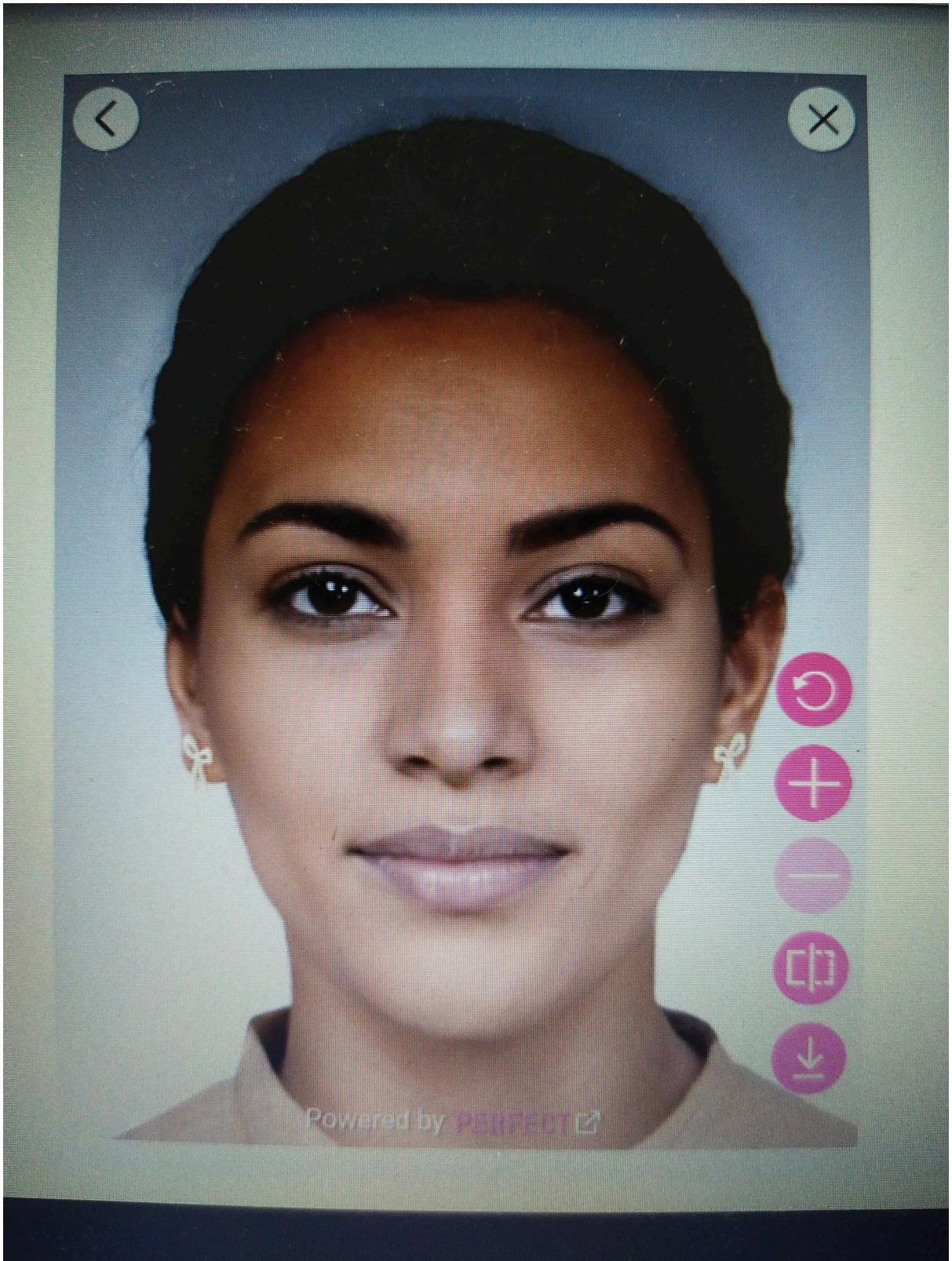
Facial and hand landmarks are detected, and jewellery models are positioned according to the detected locations.

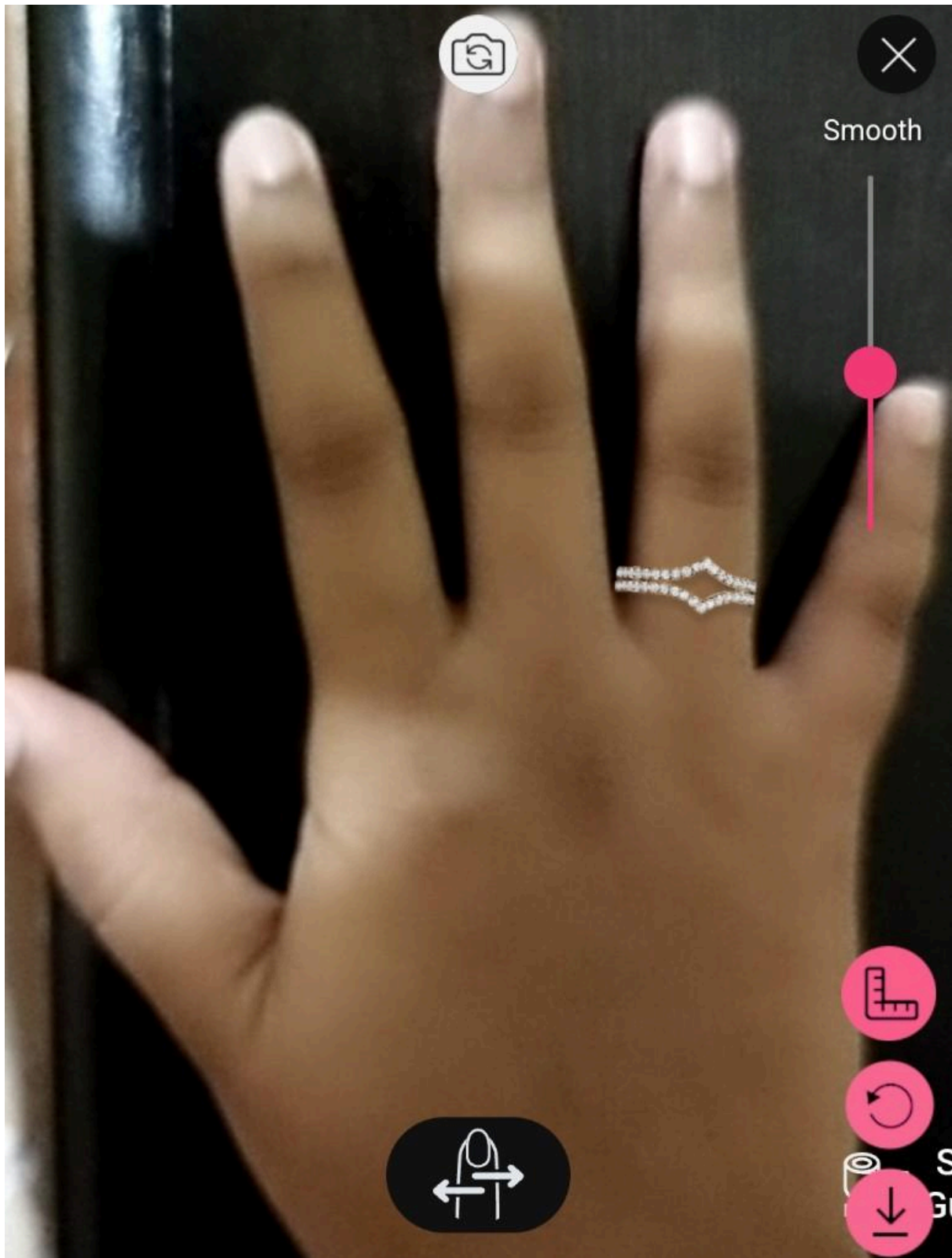
Advantages

- * Accurate placement
- * Real-time interaction
- * Improves customer confidence

Limitations

- * Occlusion issues
- * Requires precise landmark detection





4. Glasses Try-On

Technology Used

* Face Mesh

- * Facial Landmark Detection

Common Technologies

- * MediaPipe Face Mesh
- * ARKit Face Tracking

Working

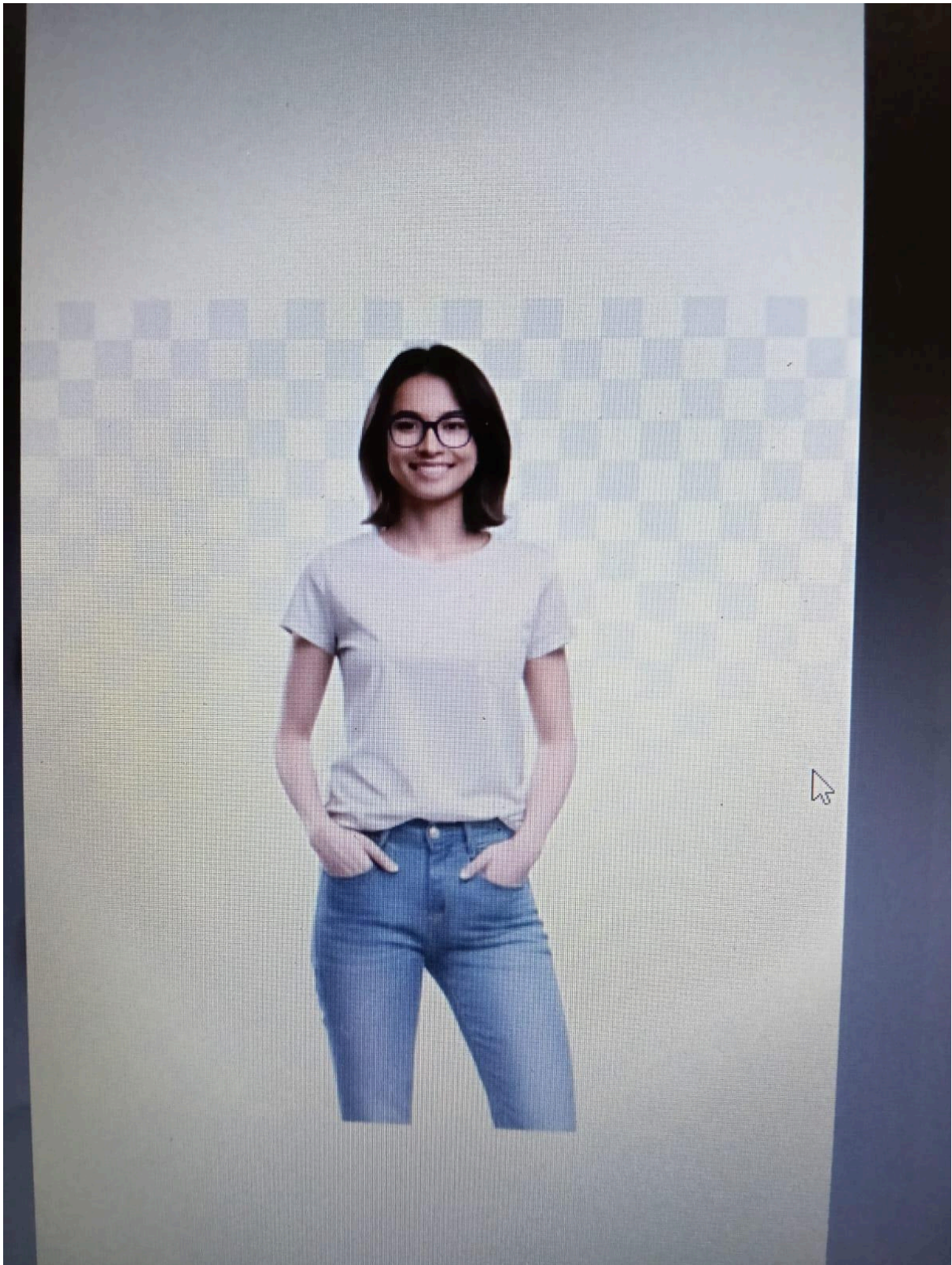
The system detects facial landmarks and places virtual glasses according to eye and nose positions.

Advantages

- * High tracking accuracy
- * Realistic fitting experience
- * Easy to use

Limitations

- * Performance affected by extreme head angles
- * Requires clear facial visibility



5. Watch Try-On

Technology Used

* Hand Tracking

- * Wrist Detection

Common Technologies

- * MediaPipe Hands
- * ARKit Hand Tracking

Working

Hand landmarks are detected, and a watch model is placed around the wrist region.

Advantages

- * Real-time placement
- * Good user engagement

Limitations

- * Tracking may fail when wrist is partially hidden



6. Hat Try-On

Technology Used

- * Head Tracking
- * Face Tracking

Working

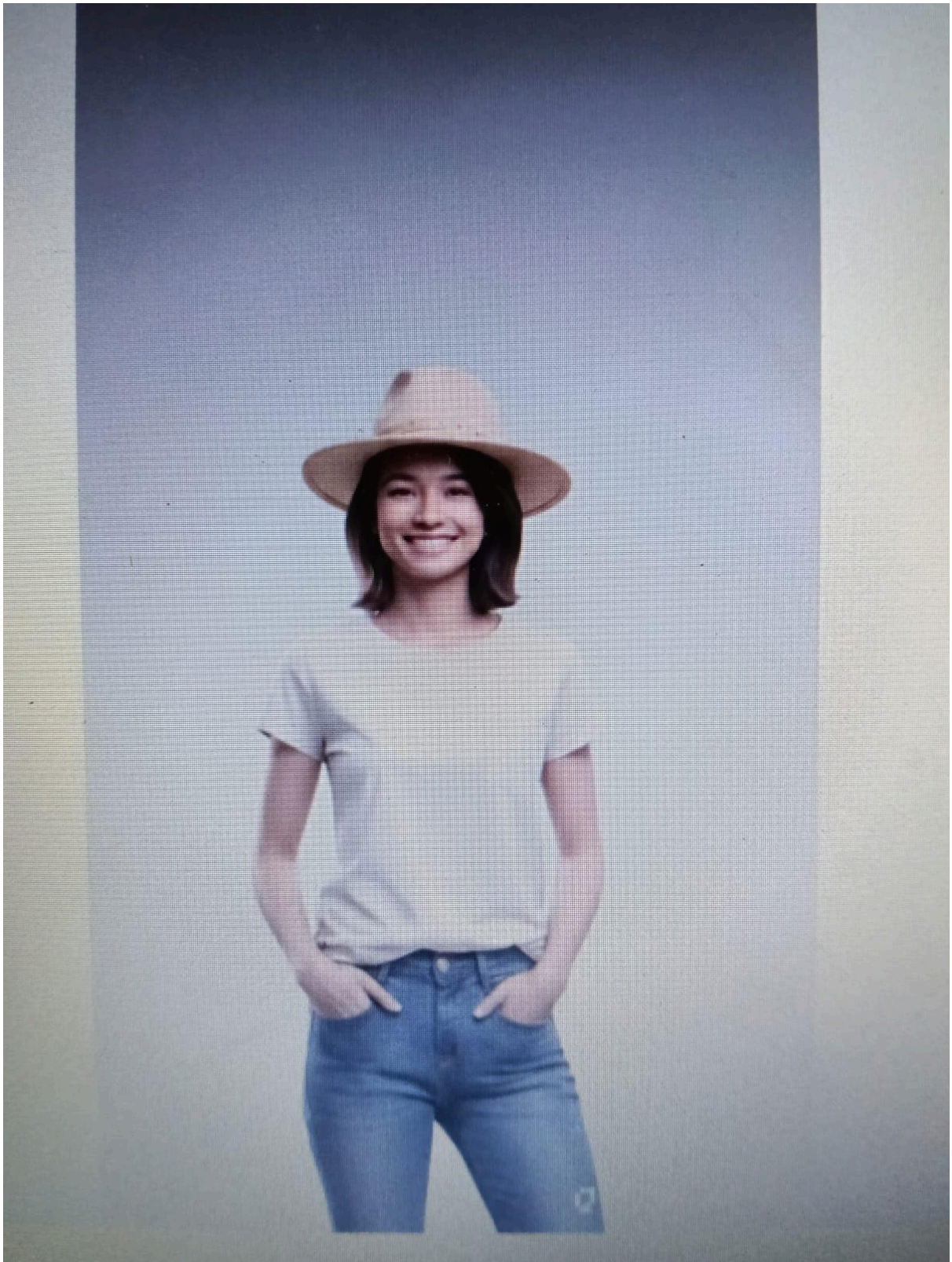
The system estimates head position and orientation to place virtual hats correctly.

Advantages

- * Easy implementation
- * Real-time interaction

Limitations

- * Positioning errors during rapid head movement



7. Bag Try-On

Technology Used

* Body Tracking

- * Shoulder Detection

Working

The bag model is aligned with the shoulder or hand position detected by the tracking system.

Advantages

- * Useful for fashion retail
- * Interactive visualization

Limitations

- * Limited realism without 3D simulations.



Category	Tracking Method	Technology Used	Accuracy	Real-Time Support
Clothes	Body Tracking	MediaPipe Pose	High	Yes
Shoes	Foot Tracking	ARCore, ARKit	High	Yes
Jewellery	Face/Wrist Tracking	Face Mesh	High	Yes
Glasses	Face Tracking	Face Mesh	Very High	Yes
Watches	Hand Tracking	MediaPipe Hands	High	Yes
Hats	Head Tracking	Face Tracking	High	Yes
Bags	Body Tracking	BlazePose	Medium-High	Yes

Final Recommendation

For developing an AR-based virtual try-on system:

- * Clothes Try-On: MediaPipe Pose + Body Segmentation
- * Shoe Try-On: ARCore or ARKit
- * Jewellery Try-On: Face Mesh + Wrist Tracking
- * Glasses Try-On: MediaPipe Face Mesh
- * Watch Try-On: MediaPipe Hands
- * Development Platform: Unity AR Foundation

These technologies provide high accuracy, real-time performance, and compatibility with modern mobile devices.

Conclusion

AR try-on technology enhances online shopping by allowing users to visualize products before purchase. Technologies such as MediaPipe, ARCore, ARKit, and Face Mesh offer reliable tracking and real-time rendering capabilities, making them suitable for implementing clothes, shoes, jewellery, and accessory virtual try-on systems.